MECHANICAL ENGINEERING
Complete Domain Learning Guide

10 modules • Engineering explanations • Worked calculations • Checkpoints • Practice assessment

A practical guide from requirements and materials to strength, CAD, thermal-fluid systems, manufacturing, quality, maintenance and safety.

Practice assessment: 20 questions
Suggested pass target: 14/20 (70%)

How to use this guide

Read the concepts, follow the worked calculations with units, and use the checkpoints to test understanding. Real engineering decisions require verified drawings, applicable standards, suitable material data and review by qualified personnel. Examples in this guide are educational, not a substitute for a design validation.

Engineering habit: Define the requirement, state assumptions, calculate with consistent units, check failure modes, and verify the result against applicable requirements.

1. The Engineering Design Process

Functional requirements

A functional requirement states what the design must do in a way that can be verified. ‘Support a 20 kg load for 10 minutes without permanent deformation’ is more testable than ‘be strong’. Requirements should be clear, measurable, traceable and prioritised.

Design constraints

Constraints are limits the solution must respect: maximum cost, envelope size, mass, temperature, noise, materials, regulations, manufacturing capability or service life. A concept that performs well but violates a hard constraint is not acceptable.

Concept generation and selection

Generate several feasible concepts before selecting one. A weighted decision matrix lists criteria, assigns weights, scores each concept and calculates weighted totals. Scores support discussion; they do not replace engineering judgement or safety requirements.

Worked selection matrix

Suppose criteria weights are strength 0.4, cost 0.3 and ease of manufacture 0.3. Concept A scores 8, 6, 7; weighted score = 0.4×8 + 0.3×6 + 0.3×7 = 7.1. Concept B scores 7, 8, 8; total = 7.6. B scores higher under these assumptions, but hard constraints and uncertainty must still be checked.

Design review and traceability

A design review checks requirements, calculations, drawings, risks, interfaces, test plans and unresolved questions before the project proceeds. Maintain a decision log showing the choice, alternatives considered, evidence, assumptions, owner and date.

Module checkpoint
What makes a functional requirement useful?

Answer: It states a measurable outcome that can be verified by test or inspection.

2. Engineering Materials

Material families

Metals often provide useful strength, toughness and manufacturability; polymers are light and corrosion-resistant but can creep or soften with heat; ceramics are hard and heat-resistant but often brittle; composites combine constituents to achieve tailored properties but may be anisotropic and harder to inspect or repair. No material is best for every use.

Stress–strain curve

Stress is load per original cross-sectional area; strain is change in length divided by original length. In the elastic region, deformation is largely recoverable. Yield strength marks the onset of significant permanent deformation. Ultimate tensile strength is the maximum engineering stress reached; fracture occurs later or near the maximum depending on the material.

Yield strength and ductility

Yield strength indicates resistance to permanent deformation. Ductility describes plastic deformation before fracture, often measured by elongation or reduction in area. High strength does not automatically mean high ductility or toughness.

Fatigue

Fatigue is progressive damage from repeated or fluctuating loads, sometimes below the static strength. Stress concentration, surface finish, corrosion, mean stress and number of cycles affect life. Rotating shafts, springs and vehicle components may need fatigue assessment, not only a one-time load check.

Corrosion resistance

Corrosion is chemical or electrochemical degradation. Selection depends on environment, moisture, salts, temperature, contact with dissimilar metals, coatings and maintenance. Stainless steel is not immune to every corrosion mode; galvanic corrosion can occur when dissimilar metals contact in an electrolyte.

Material selection scenario

For a lightweight outdoor bracket, compare strength-to-weight, fatigue, corrosion, joining method, cost and temperature. An aluminium alloy, coated steel or composite may be plausible depending on load and environment; select only after checking design requirements and validated material data.

Module checkpoint
A repeated cyclic load can cause which failure mechanism even below static strength?

Answer: Fatigue.

3. Statics & Strength of Materials

Force and equilibrium

A body in static equilibrium has net force and net moment equal to zero. Resolve forces into components, draw a free-body diagram, identify supports and loads, then apply $\Sigma F_x=0$, $\Sigma F_y=0$ and $\Sigma M=0$. Choose a consistent sign convention.

Stress and strain

For a straight member under axial load, average normal stress $\sigma = F/A$. Engineering strain $\varepsilon = \Delta L/L$■. In the linear elastic range, $\sigma = E\varepsilon$, where E is Young's modulus. These simple relations assume suitable geometry, material behaviour and loading.

Worked axial stress

A bar carries 12 kN in tension and has a cross-sectional area of 300 mm². Average stress = 12,000 N ÷ 300 mm² = 40 N/mm² = 40 MPa, because 1 N/mm² = 1 MPa. This is not by itself proof that the bar is safe; compare against appropriate design allowables and failure modes.

Bending moment

Bending moment is the internal moment at a beam section. For a simply supported beam with a central point load P and span L, the maximum bending moment is PL/4. For a cantilever with end load P, the maximum moment at the fixed support is PL. Use the formula matching the support and load arrangement.

Bending stress and deflection

For elastic beam bending, normal stress varies with distance from the neutral axis; a common relation is $\sigma = My/I$, where M is bending moment, y is distance from the neutral axis and I is second moment of area. Deflection depends on load, span, stiffness and support conditions.

Factor of safety

A simplified factor of safety is material strength or design capacity divided by the relevant applied stress or load effect. The correct value and basis depend on failure mode, material variability, loading, consequence and applicable code. It does not compensate for incorrect assumptions or missing load cases.

Module checkpoint
A 10 kN force acts on 200 mm². What is average axial stress?

Answer: 10,000 N ÷ 200 mm² = 50 N/mm² = 50 MPa.

4. CAD & Technical Drawing

Parametric modelling

A parametric model is driven by dimensions, constraints and relationships. If a hole must remain centred as a plate width changes, constrain its position relative to the plate rather than placing it by eye. Fully constrained sketches reduce accidental geometry changes.

Orthographic projection

Orthographic drawings use aligned 2D views such as front, top and side to communicate 3D geometry. First-angle and third-angle projection place views differently, so the projection method must be identified. Sections reveal internal features that hidden lines may make difficult to interpret.

Dimensioning

Dimension functional geometry clearly and avoid duplicate or conflicting dimensions. Include units, datums, surface finish and other requirements where needed. Dimensions should allow manufacture and inspection without forcing a shop to infer design intent.

Tolerance

Tolerance is the permissible variation around a nominal dimension. For 20.00 ±0.10 mm, acceptable limits are 19.90 to 20.10 mm. Tighter tolerances often increase machining, inspection and scrap costs; specify them where function requires them.

Fits and assembly constraints

Clearance fits allow relative movement; interference fits rely on elastic deformation and contact pressure; transition fits may produce either small clearance or interference. CAD assembly constraints define how components align or move. Confirm the model reflects intended physical degrees of freedom.

Scenario

A shaft must rotate in a bearing. The drawing should define shaft and bore sizes, fit/tolerance, material, finish and relevant datum relationships. A visually correct CAD assembly is not enough if the tolerances do not support function.

> **Module checkpoint**
> For a 20.00 ±0.10 mm dimension, what are the limits?

Answer: 19.90 mm and 20.10 mm.

5. Thermal & Fluid Fundamentals

Conduction

Conduction transfers heat through a material without bulk movement. For steady one-dimensional conduction through a flat wall, Q■ = kAΔT/L, where k is thermal conductivity, A is area, ΔT is temperature difference and L is thickness. The relation assumes a simplified geometry and steady conditions.

Convection and radiation

Convection transfers heat between a surface and a moving fluid; a common estimate is Q■ = hA(Ts−T∞). Radiation transfers energy by electromagnetic waves and does not require a material medium. Real systems can combine all three modes.

Worked conduction example

A wall has k=0.8 W/(m·K), area 2 m², thickness 0.1 m and temperature difference 15 K. Ideal steady conduction rate = 0.8×2×15/0.1 = 240 W. Contact resistance, layers and changing conditions could alter actual performance.

Laminar and turbulent flow

Laminar flow is relatively orderly, with limited mixing between layers; turbulent flow contains fluctuating motion and stronger mixing. Reynolds number Re = ρvD/μ for internal flow in a pipe, where ρ is density, v average velocity, D diameter and μ dynamic viscosity. Transition depends on geometry and disturbance; the familiar pipe-flow thresholds are approximate.

Pressure drop

Pressure is lost along pipes due to wall friction and fittings. Darcy–Weisbach gives ΔP = f(L/D)(ρv²/2) for a straight pipe, with additional local losses often represented as K(ρv²/2). The friction factor depends on Reynolds number and relative roughness.

Sizing considerations

A flow or thermal element must meet required capacity without unacceptable pressure drop, temperature rise, noise, energy use or safety risk. Use appropriate fluid properties and boundary conditions; confirm with test data or validated calculations.

> **Module checkpoint**
> Which heat transfer mode occurs through a solid without bulk material movement?

Answer: Conduction.

6. Manufacturing Processes

Casting

Molten material is poured or injected into a mould and solidifies. Casting can produce complex shapes and can be economical at suitable volumes, but shrinkage, porosity, inclusions and surface finish require process control and inspection.

Machining

Machining removes material with cutting tools. It is versatile and can achieve good dimensional accuracy, but generates chips and may have higher material waste and cycle time for complex or high-volume parts. Tool wear, fixturing and cutting conditions affect quality.

Forming

Processes such as forging, rolling, extrusion and sheet-metal forming change shape through plastic deformation. They can provide good material utilisation and production rates but need suitable tooling, material ductility and process design.

Additive manufacturing

Additive manufacturing builds parts layer by layer from digital data. It is useful for complex geometries, custom parts and prototypes, but orientation, support structures, anisotropy, surface finish, material qualification and post-processing matter.

Tooling cost and production volume

Tooling cost is an upfront investment in moulds, dies, jigs or fixtures. A simplified unit-cost model is variable cost per part + tooling cost/volume. For example, ■5,00,000 tooling spread across 10,000 parts adds ■50 per part before other costs; spread across 500 parts it adds ■1,000 per part.

Process selection

Choose by material, geometry, tolerance, surface finish, volume, lead time, tooling budget, scrap and inspection needs. A process that is excellent for one prototype may be uneconomical for mass production.

Module checkpoint
Why can tooling make a process expensive at low volume?

Answer: The upfront tooling cost is spread over fewer parts, increasing tooling cost per part.

7. Design for Manufacturability

Design for Manufacturability (DFM)

DFM makes a design easier and more economical to produce using the intended process. Reduce unnecessary features, use accessible tool paths, avoid awkward setups, standardise sizes and design for inspection and assembly as well as production.

Draft angle

A draft angle is a slight taper that helps a moulded or cast part release from the tool. Required draft depends on process, material, surface texture, depth and tool design. Insufficient draft can cause sticking, scuffing or damage.

Standardisation

Using common fasteners, hole sizes, bearings, materials and features can reduce purchasing complexity, inventory, tooling and errors. Standardisation should not override a functional or safety requirement.

Part consolidation

Combining parts can reduce fasteners, assembly time and potential failure points. However, it can make repair harder, increase tooling complexity or force replacement of a larger assembly after a local failure. Evaluate lifecycle cost and serviceability.

Tolerance versus cost

Use tight tolerances only where required for fit, function or safety. A tolerance stack-up occurs when variations across multiple components accumulate in an assembly. Analyse critical dimensions and consider datum schemes and statistical variation where appropriate.

Scenario

A bracket has five unnecessary machined pockets and four unique fastener sizes. Simplifying the geometry and using a standard fastener family may reduce machining time and assembly errors without compromising strength.

Module checkpoint
What is the purpose of draft angle in moulding or casting?

Answer: To help the part release from the tool.

8. Quality & Metrology

Metrology and instrument choice

Metrology is the science of measurement. Choose an instrument based on dimension, tolerance, range, resolution, accuracy, repeatability and environment. A ruler is unsuitable for a tight bore tolerance; a calibrated bore gauge or suitable internal micrometer may be required.

Calibration and uncertainty

Calibration relates instrument readings to known standards and documents error. Resolution is the smallest displayed increment, while accuracy concerns closeness to the true value. Repeatability describes agreement under repeated conditions. No instrument produces perfectly certain measurements.

Statistical Process Control (SPC)

SPC uses process data over time and control charts to distinguish common-cause variation from signals of special causes. Control limits are calculated from process behaviour; specification limits come from design requirements. A process can be statistically stable yet still produce parts outside specification.

Process capability

Cp compares tolerance width with process spread, commonly 6σ: $C_p=(USL-LSL)/(6\sigma)$. Cpk also accounts for how centred the process mean is: $C_{pk}=\min(USL-\mu, \mu-LSL)/(3\sigma)$. These indices require appropriate data and assumptions; capability is not meaningful if the process is unstable.

Worked capability example

For USL=10.5 mm, LSL=9.5 mm and σ=0.10 mm, $C_p=(1.0)/(0.6)=1.67$. If mean μ=10.0 mm, $C_{pk}=\min(0.5,0.5)/(0.3)=1.67$. If the mean shifts to 10.2 mm, $C_{pk}=\min(0.3,0.7)/0.3=1.00$, while Cp remains 1.67.

Non-conformance

A non-conformance fails a specified requirement. Contain affected parts, identify scope and traceability, document evidence, assess risk, determine disposition through authorised procedures and investigate cause. Do not silently rework or accept a defect without approval.

Module checkpoint
How do control limits differ from specification limits?

Answer: Control limits reflect process behaviour; specification limits come from design requirements.

9. Maintenance & Reliability

Preventive, predictive and corrective maintenance

Preventive maintenance occurs on a schedule to reduce failure likelihood. Predictive maintenance uses condition data—such as vibration, temperature, oil analysis or current—to indicate when service is needed. Corrective maintenance restores equipment after a fault; it may be planned or unplanned.

Condition indicators

Rising vibration may indicate imbalance, misalignment, looseness or bearing damage; temperature changes can indicate friction or lubrication problems. A single signal rarely proves a specific cause. Establish baselines, trend measurements and use trained diagnosis.

Root cause failure analysis

Root cause analysis asks why a failure occurred rather than only replacing the failed component. Tools include the 5 Whys, fishbone diagrams, fault trees and failure mode analysis. Confirm causes with evidence and define corrective actions that prevent recurrence.

Downtime and availability

Downtime is time equipment is unavailable due to failure or maintenance. A simplified availability measure is uptime ÷ (uptime + downtime). If a machine is available for 450 of 480 scheduled minutes, availability is 93.75%. Define whether planned maintenance is included.

Criticality-based planning

Prioritise equipment by safety consequences, environmental impact, production loss, repair lead time and redundancy. Critical assets may need condition monitoring and spare-parts planning; low-criticality equipment may use simpler scheduled checks.

Scenario

A pump repeatedly fails after its bearing is replaced. Root cause analysis should check alignment, lubrication, contamination, operating conditions and installation procedure rather than repeatedly changing the bearing alone.

Module checkpoint
Which maintenance type uses measured condition indicators to decide when service is needed?

Answer: Predictive maintenance.

10. Safety & Standards

Risk assessment

Identify hazards, estimate likelihood and severity, evaluate risk and choose controls. Consider normal operation, setup, cleaning, maintenance, foreseeable misuse and failure conditions. Reassess when the design, process or operating environment changes.

Hierarchy of controls

Prefer elimination, then substitution, engineering controls, administrative controls and personal protective equipment. PPE is important but usually less reliable as the sole control because it depends on correct selection and use.

Engineering standards

Standards define technical requirements, test methods, dimensions, safety practices or quality procedures. Identify the correct edition and scope, verify whether it is legally or contractually applicable, and document the standard and clause used. Never cite a standard from memory if the exact requirement matters.

Guards and interlocks

Guards physically prevent access to dangerous moving parts. An interlock prevents or stops operation when a defined safe condition is not met, such as a guard being open. Interlocks should be designed for foreseeable faults and must not be easily bypassed; they are not a substitute for risk assessment.

Safe design review

Check pinch points, rotating shafts, stored energy, hot surfaces, pressure, lifting, electrical interfaces, access for maintenance, emergency stopping and failure modes. Validate safety-related functions to the required level using applicable standards and competent review.

Scenario

A rotating shaft is accessible while a machine runs. The preferred response is to assess the hazard and use suitable guarding and, where necessary, a properly designed interlock—not merely place a warning sticker.

Module checkpoint
What is the preferred first level in the hierarchy of controls?

Answer: Elimination of the hazard, where feasible.

Cross-module worked examples

1. Axial stress and factor of safety

A component carries 8,000 N over a 400 mm² area. Average stress = 8,000/400 = 20 MPa. If a simplified allowable stress is 80 MPa, the ratio allowable/applied is 80/20 = 4. This is an illustrative calculation only; real design must use the correct load combinations, failure criteria, material data and applicable code.

2. Manufacturing unit cost

A mould costs ■3,60,000 and variable manufacturing cost is ■120 per part. At 6,000 parts, tooling adds ■60 per part, so simplified cost is ■180 per part. At 600 parts, tooling adds ■600 per part, so simplified cost is ■720 per part. Excludes labour, overhead, scrap, inspection and other costs.

3. Quality capability

With USL 25.2 mm, LSL 24.8 mm and standard deviation 0.05 mm, Cp = 0.4/(6×0.05)=1.33. If the mean is 25.1 mm, Cpk = min(0.1,0.3)/(3×0.05)=0.67. The spread fits the tolerance width in theory, but the off-centre process is at risk on the upper side.

4. Maintenance availability

If a machine has 72 hours of uptime and 8 hours of downtime during a defined period, availability = 72/(72+8) × 100% = 90%. State whether planned maintenance is included and use a consistent reporting window.

Practice assessment — 20 questions

Choose one best answer for each question. The answer key includes explanations. A score of 14/20 equals 70%.

1. A good functional requirement should be:

A. Vague and inspirational

B. Measurable and verifiable

C. Only about appearance

D. Impossible to test

2. A design constraint is:

A. A fixed limit such as size or cost

B. A marketing slogan

C. A test result only

D. A material defect

3. Yield strength is the stress at which:

A. A material begins permanent deformation

B. A part becomes corrosion-proof

C. Heat transfer stops

D. A drawing is approved

4. Fatigue damage is associated with:

A. Only one static load

B. Repeated or fluctuating loading

C. CAD constraints

D. Tooling invoices

5. A 6 kN force acts on 150 mm². Average stress is:

A. 4 MPa

B. 40 MPa

C. 400 MPa

D. 900 MPa

6. Strain is:

A. Force per area

B. Change in length divided by original length

C. Bending moment times area

D. Heat per second

7. For a 12.0 ±0.2 mm dimension, acceptable limits are:

A. 11.8 to 12.2 mm

B. 12.0 to 12.2 mm

C. 11.0 to 12.0 mm

D. 11.2 to 12.8 mm

8. Which mode transfers heat through a solid by molecular interaction?

A. Conduction

B. Convection

C. Advection only

D. Radiation only

9. Which statement about pressure drop is correct?

A. It is always zero in a pipe

B. Friction and fittings can reduce pressure along a flow path

C. It is the same as temperature

D. It depends only on pipe colour

10. Which process builds a part layer by layer from digital data?

A. Machining

B. Casting

C. Additive manufacturing

D. Forging only

11. Tooling costs per part generally decrease when:

A. Production volume increases

B. Production volume decreases

C. The mould is discarded before use

D. Every part uses unique tooling

12. Draft angle helps:

A. Parts release from moulds or dies

B. Increase electrical voltage

C. Calculate CTR

D. Measure customer churn

13. Which is a benefit of standardisation?

A. More unique parts to stock

B. Reduced inventory complexity and fewer variants

C. Elimination of all design checks

D. Guaranteed zero defects

14. Control limits in SPC are based mainly on:

A. Observed process behaviour

B. The customer's advertising budget

C. A random nominal value

D. Tool colour

15. A non-conformance means:

A. A product fails a specified requirement

B. A product has a new feature

C. A machine is always available

D. A drawing has three views

16. Maintenance triggered by vibration or temperature trends is:

A. Predictive

B. Corrective only

C. Unplanned by definition

D. Design review

17. Root cause analysis aims to:

A. Replace a failed part repeatedly

B. Identify underlying causes and prevent recurrence

C. Hide defects

D. Increase downtime

18. The first preferred control in the hierarchy is:

A. PPE

B. Warning sign

C. Elimination

D. Training only

19. An interlock is intended to:

A. Prevent or stop operation unless required conditions are met

B. Increase tool wear

C. Replace every guard

D. Set the product price

20. Which statement about a factor of safety is best?

A. It removes the need for load analysis

B. It compares capacity with demand on a defined basis but does not replace sound design

C. It guarantees no failure under every condition

D. It is the same as tolerance

Answer key with explanations

1. B. Verification makes it possible to check whether the design meets the need.

2. A. Constraints bound acceptable solutions.

3. A. Yield marks the onset of plastic deformation.

4. B. Fatigue can accumulate under cyclic loads.

5. B. 6,000/150 = 40 N/mm² = 40 MPa.

6. B. Strain is a dimensionless measure of deformation.

7. A. Subtract and add the tolerance to the nominal dimension.

8. A. Conduction does not require bulk material motion.

9. B. Flow resistance produces pressure losses.

10. C. Additive processes build material in layers.

11. A. A fixed upfront cost is distributed across more units.

12. A. Taper can reduce sticking and damage during ejection.

13. B. Common components can reduce cost and error.

14. A. Control limits help identify unusual process variation.

15. A. It requires containment and controlled disposition.

16. A. Condition indicators inform maintenance timing.

17. B. It seeks causes rather than only symptoms.

18. C. Elimination removes the hazard rather than relying on later controls.

19. A. Interlocks link machine operation to defined safety conditions.

20. B. Safety factors must be used with correct assumptions, failure modes and standards.

Quick revision sheet

Term	Remember
Functional requirement	Measurable statement of what the design must achieve.
Constraint	Hard limit such as cost, size, mass or regulation.
Yield strength	Stress at onset of permanent deformation.
Ductility	Plastic deformation before fracture.
Fatigue	Progressive damage under cyclic loading.
Stress	Axial average stress $\sigma = F/A$.
Strain	$\varepsilon = \Delta L/L$■.
Elastic relation	$\sigma = E\varepsilon$ in the linear elastic region.
Bending stress	$\sigma = My/I$ for simplified elastic beam bending.
Factor of safety	Capacity divided by demand on a defined basis.
Parametric CAD	Dimensions and constraints drive geometry.
Tolerance	Permissible variation around a nominal dimension.
Conduction	Heat transfer through material without bulk motion.
Convection	Heat transfer between a surface and moving fluid.
Reynolds number	$Re = \rho vD/\mu$ for internal pipe flow.
Pressure drop	Loss from friction and fittings.
Casting / machining / additive	Mould and solidify / remove material / build layer by layer.
DFM	Design geometry and features for economical manufacture.
SPC	Monitor process variation over time.
Cp / Cpk	Capability spread index / capability accounting for centring.
Preventive / predictive	Scheduled service / condition-triggered service.
Root cause analysis	Find underlying reasons and prevent recurrence.
Hierarchy of controls	Elimination, substitution, engineering, administrative, PPE.
Guard / interlock	Physical access barrier / control tied to safe conditions.

Exam and practice tip: Draw a free-body diagram for mechanics questions, write units at every calculation step, check assumptions, and never treat an illustrative calculation as approval for a real load-bearing or safety-critical design.